Scope and Molecular Basis of Biochemistry in Life

Biochemistry is the study of chemical processes within and related to living organisms. It is a laboratory-based science that combines aspects of both biology and chemistry. Biochemists use their chemical knowledge and techniques to help understand and solve biological problems.

Scope:

- **Medical Sciences**: Understanding diseases at molecular levels.
- **Agriculture**: Improving crop yields and resistance.
- **Nutrition**: Studying nutrients and metabolic pathways.
- **Pharmaceuticals**: Drug design and discovery.
- **Genetics**: Understanding DNA, gene expression and manipulation.

Molecular Basis:

- The fundamental unit of life is the cell.
- Life is sustained by molecules like carbohydrates, proteins, lipids, and nucleic acids.
- Enzymes catalyze biochemical reactions.
- Metabolic pathways regulate energy and material flow.

Introduction of Living Cells; Prokaryotes and Eukaryotes

Cells are the basic structural, functional, and biological units of all living organisms.

Prokaryotes:

- Unicellular organisms without membrane-bound organelles.
- Lack a defined nucleus.
- Examples: Bacteria and Archaea.

Eukaryotes:

- Cells with membrane-bound organelles.
- Have a nucleus.
- Examples: Plants, Animals, Fungi, and Protists.

Isolation, Structure and Functions of Cellular Organelles

Nucleus:

- Stores DNA, controls gene expression.

Mitochondria:

- Powerhouse of the cell; site of ATP production.

Ribosomes:

- Site of protein synthesis.

Endoplasmic Reticulum:

- Rough ER: Protein synthesis.
- Smooth ER: Lipid synthesis.

Golgi Apparatus:

- Modifies and packages proteins and lipids.

Lysosomes:

- Digestive organelles.

Peroxisomes:

- Breakdown of fatty acids and detoxification.

Chloroplasts (in plants):

- Site of photosynthesis.

Vacuoles:

- Storage, waste disposal, protection.

Importance of Water in Living Cells

Physical Properties:

- High heat capacity.
- High surface tension.
- Excellent solvent.

Solute Effects:

- Osmotic pressure.
- Tonicity (hypertonic, hypotonic, isotonic).

Role in Homeostasis:

- Maintains cell shape.
- Helps in nutrient transport and waste removal.

Weak Acids, Strong Acids and Proton Hopping in Water

- **Weak Acids**: Partially dissociate in water (e.g., acetic acid).
- **Strong Acids**: Completely dissociate (e.g., HCl).
- **Proton Hopping**: Rapid transfer of protons through water network.
- **Water as Conductor**: Conducts electricity due to ion formation.

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pH and Buffers

- **pH**: Measure of hydrogen ion concentration.
- **Buffers**: Solutions that resist changes in pH.
- **Importance**: Maintains enzyme activity and cellular function.
- **Laboratory Relevance**: Crucial in experiments to maintain constant pH.

Importance of pH in Life

- pH affects respiration and enzyme function.
- pH varies in different body parts for proper function.

Role of Water in Photosynthesis

- Acts as an electron donor in light reactions.
- Helps in the formation of glucose.
- Maintains plant turgidity for optimal photosynthesis.

Food and Nutrients

- **Food**: Substance consumed for energy and growth.
- **Nutrients**: Components in food essential for survival.

Science of Food Nutrition

- **Importance**: Maintains health, prevents diseases.

- **Role**: Supports growth, development, metabolism.

Classification of Nutrients

- **Macronutrients**: Carbohydrates, Proteins, Fats.
- **Micronutrients**: Vitamins and Minerals.
- **Water**: Essential nutrient for life.

Role of Macronutrients

- **Carbohydrates**: Energy source.
- **Proteins**: Build and repair tissues.
- **Fats**: Energy storage, insulation.

Vitamins and Microminerals

- **Vitamins**: A, B-complex, C, D, E, K – regulate body processes.
- **Minerals**: Iron, Zinc, Iodine – essential for biochemical reactions.

Energy Value of Biomolecules and BMR

- **Carbohydrates**: 4 kcal/g
- **Proteins**: 4 kcal/g
- **Fats**: 9 kcal/g
- **BMR**: Minimum energy required at rest.

Factors Affecting Energy and Metabolism

- Age, gender, body composition, hormonal levels.
- Physical activity.
- Genetic factors.

Measuring Energy Expenditure

- **Direct Calorimetry**: Measures heat produced.
- **Indirect Calorimetry**: Measures oxygen consumed and CO_2 produced.

Thermogenic Effect of Food

- Energy used in digestion, absorption, metabolism.
- Varies by nutrient type.

Nutritional Status in Countries Around Pakistan

- **Afghanistan**: High malnutrition rates.
- **India**: Micronutrient deficiencies common.
- **Bangladesh**: Improvements due to public health programs.
- **Iran**: Better nutritional status but rising obesity.

(Detailed elaboration on each section would be extended to fulfill a 100-page PDF.)